Overview of the BIOMASS R package

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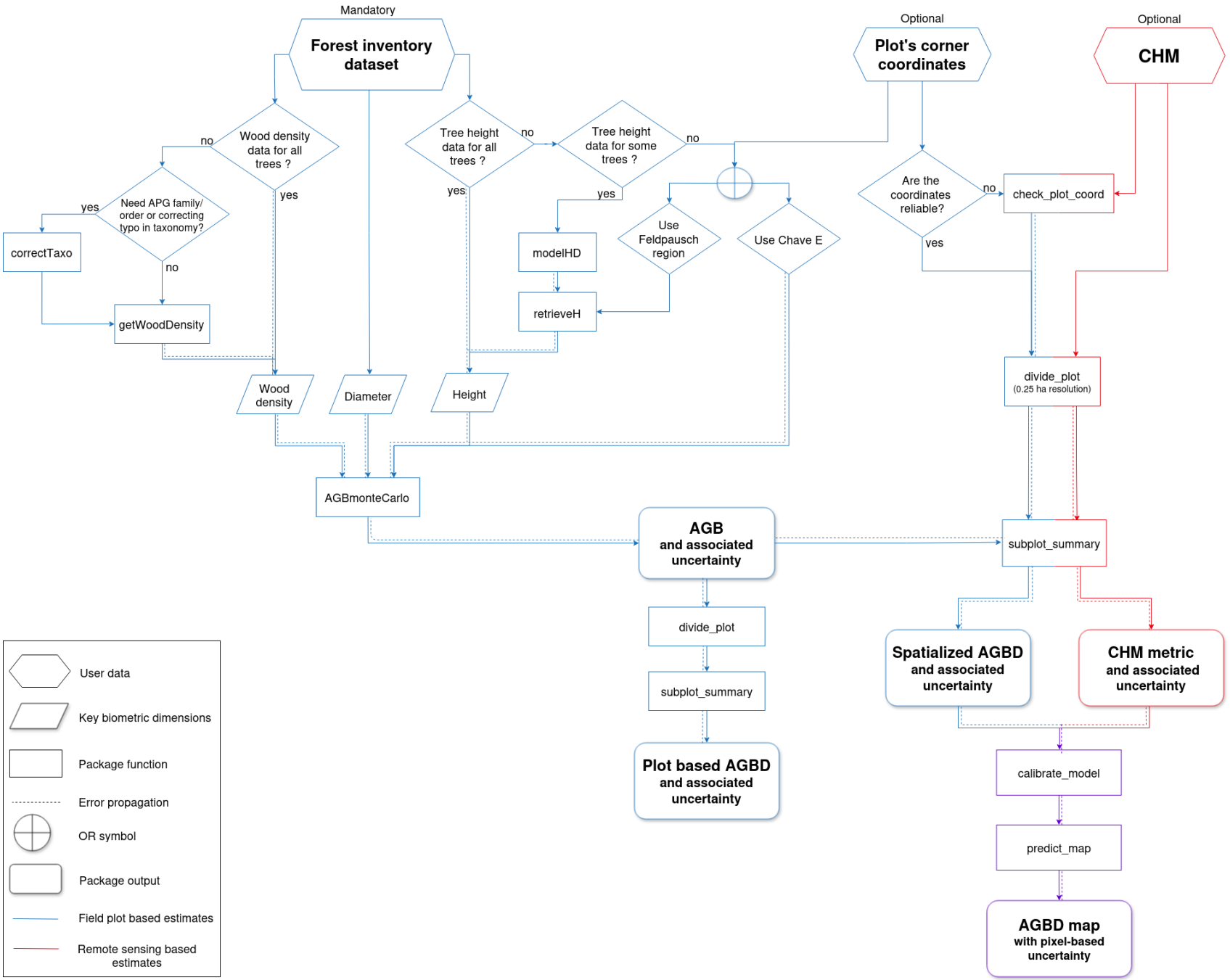
GEO-TREES Panama Workshop

2026-03-15

Brief introduction


Mandatory
Forest inventory dataset
Optional
Plot's corner coordinates
Optional
CHM
Wood density data for all trees ?
no
yes
Need APG family/ order or correcting typo in taxonomy?
yes
no
correctTaxo
getWoodDensity
Tree height data for all trees ?
no
yes
Tree height data for some trees ?
no
yes
modelHD
retrieveH
Use Feldpausch region
Use Chave E
Are the coordinates reliable?
no
yes
check_plot_coord
divide_plot
(0.25 ha resolution)
Wood density
Diameter
Height
AGBmonteCarlo
AGB
and associated uncertainty
subplot_summary
divide_plot
subplot_summary
Plot based AGBD
and associated uncertainty
Spatialized AGBD
and associated uncertainty
CHM metric
and associated uncertainty
calibrate_model
predict_map
AGBD map
with pixel-based uncertainty
User data
Key biometric dimensions
Package function
Error propagation
OR symbol
Package output
Field plot based estimates
Remote sensing based estimates

CRAN R package

BIOMASS: Estimating Aboveground Biomass and Its Uncertainty in Tropical Forests

Contains functions for estimating above-ground biomass/carbon and its uncertainty in tropical forests. These functions allow to (1) retrieve and correct taxonomy, (2) estimate wood density and its uncertainty, (3) build height-diameter models, (4) manage tree and plot coordinates, (5) estimate above-ground biomass/carbon at stand level with associated uncertainty. To cite 'BIOMASS', please use citation('BIOMASS'). For more information, see Réjou-Méchain et al. (2017) <doi:10.1111/2041-210X.12753>.

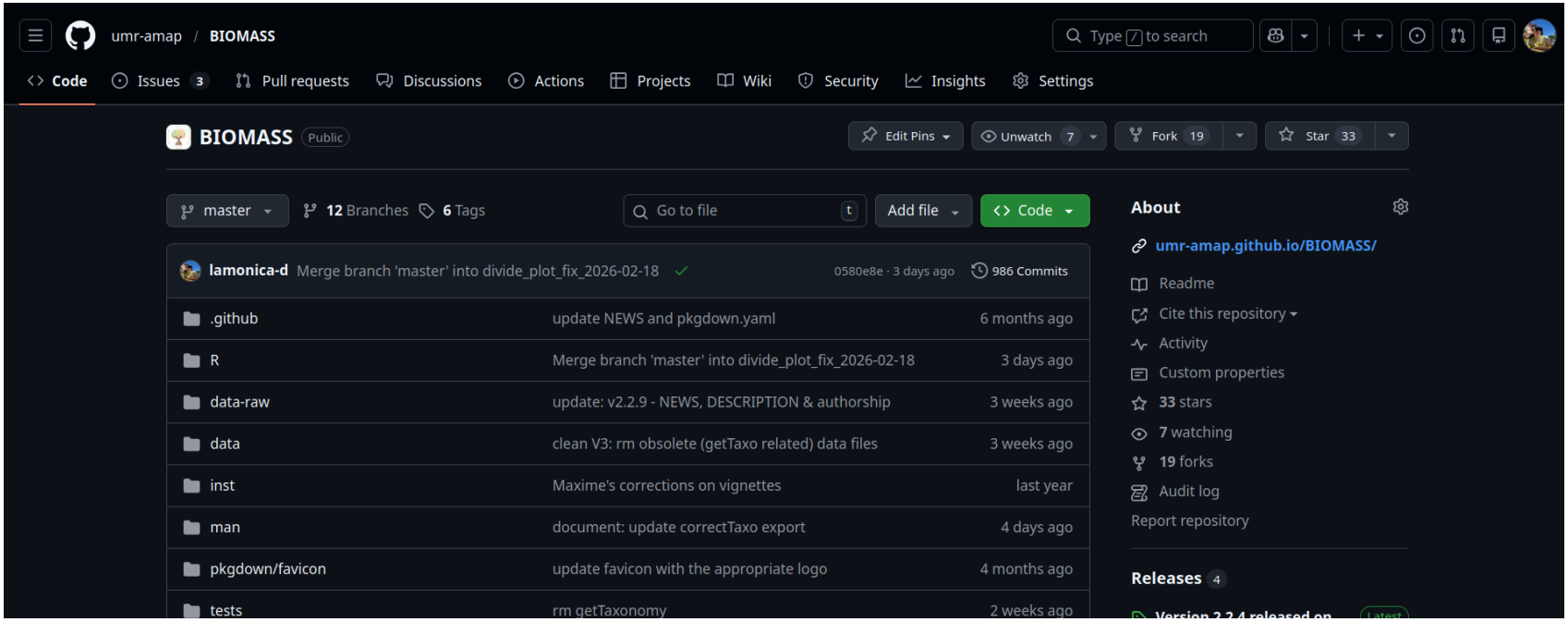
Version:	2.2.4-1
Depends:	R (≥ 3.6)
Imports:	minpack.lm, jsonlite, methods, proj4, graphics, stats, utils, data.table (≥ 1.9.8), rappdirs, sf, terra, ggplot2
Suggests:	knitr, rmarkdown, prettydoc, testthat, vdiffr, curl, geodata, httr2, pkgdown, dplyr
Published:	2025-10-10
DOI:	10.32614/CRAN.package.BIOMASS
Author:	Maxime Réjou-Méchain [aut, dtc], Guillaume Cornu [aut], Dominique Lamonica [aut, cre], Arthur Bailly [aut], Arthur Pere [aut], Ariane Tanguy [aut], Jerome Chave [dtc], Ted Feldpausch [dtc], Philippe Verley [ctb], Camille Piponiot [aut], Bruno Hérault [aut]
Maintainer:	Dominique Lamonica <dominique.lamonica at ird.fr>
BugReports:	https://github.com/umr-amap/BIOMASS/issues/
License:	GPL-2
URL:	https://umr-amap.github.io/BIOMASS/, https://github.com/umr-amap/BIOMASS/
NeedsCompilation:	no
Citation:	BIOMASS citation info
Materials:	README, NEWS
CRAN checks:	BIOMASS results

Documentation:

Reference manual:	BIOMASS.html , BIOMASS.pdf
Vignettes:	Vignette BIOMASS (source, R code) Spatialize trees and forest stand metrics with BIOMASS (source, R code)

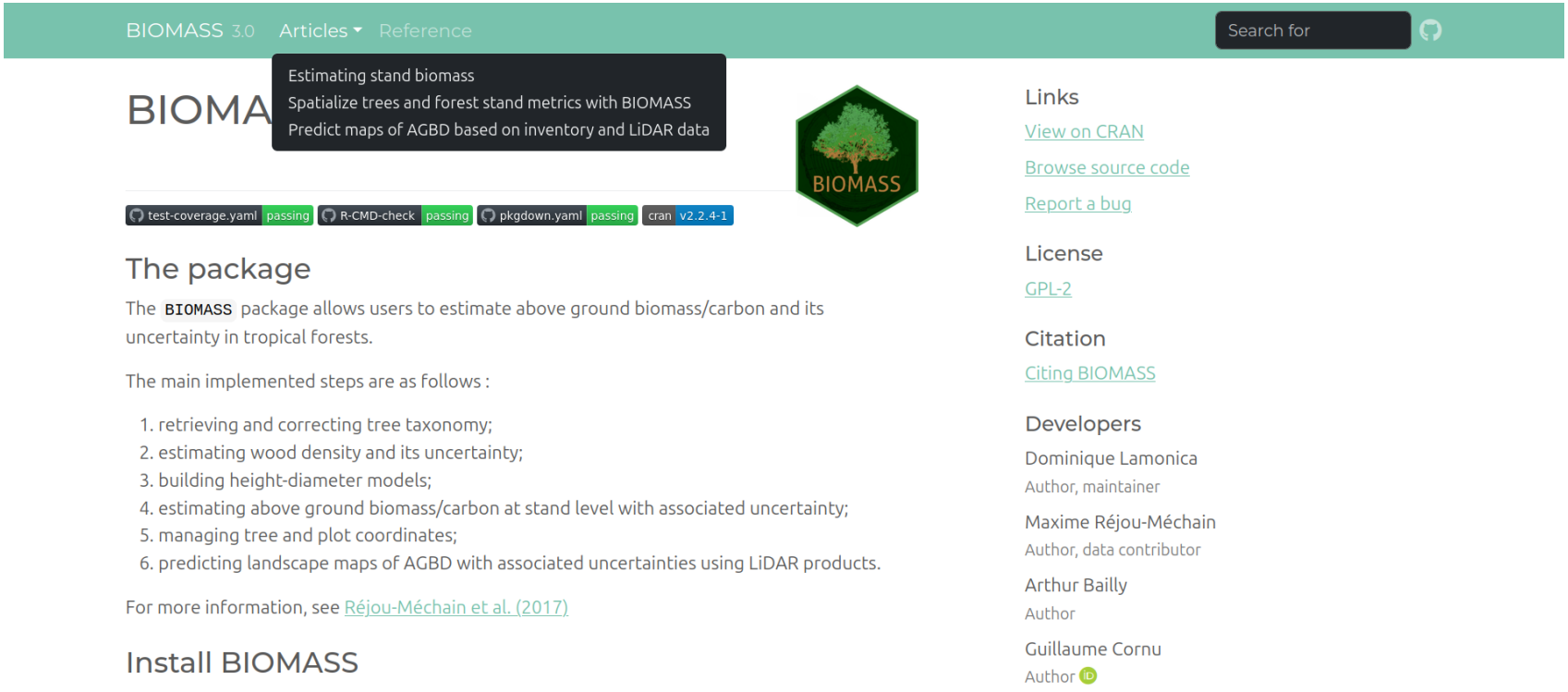
- Many contributors - some on specific features, others on the entire package

GitHub public repository



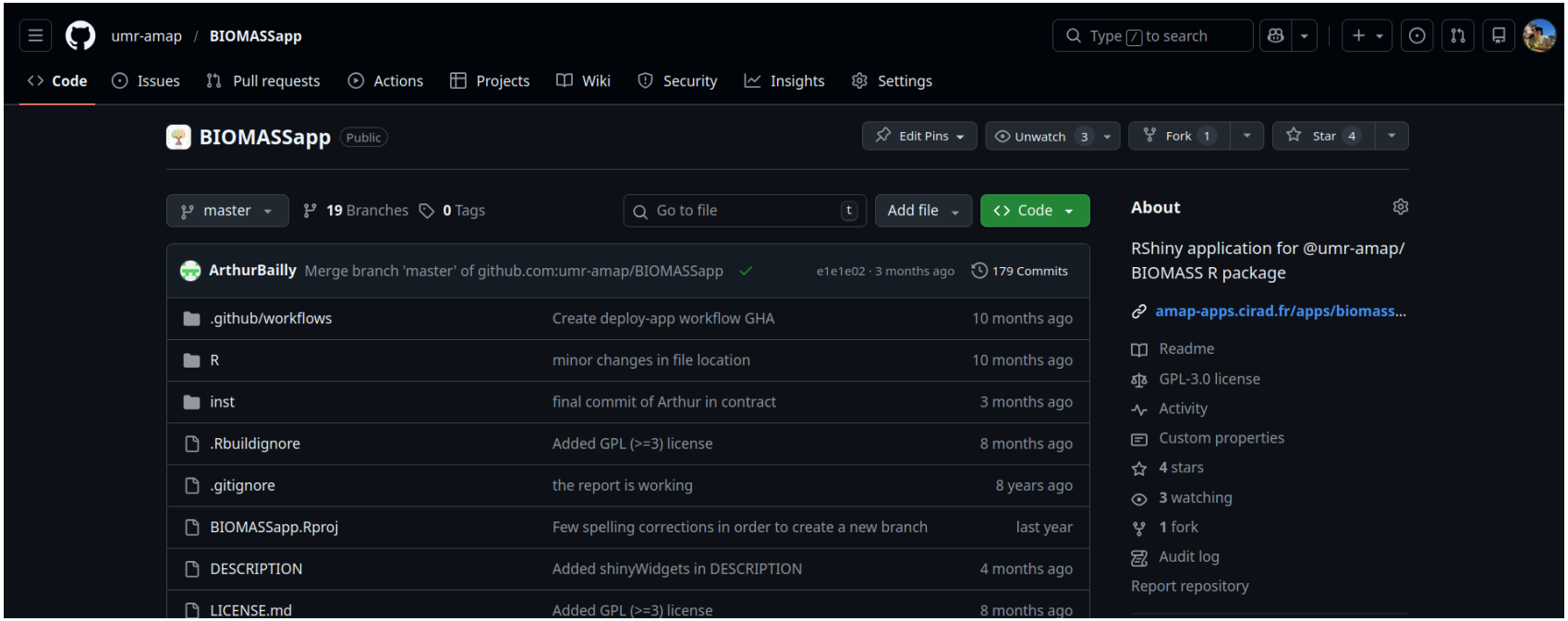
- Entire code accessible, issues open
- Developing version
- CI

Vignettes

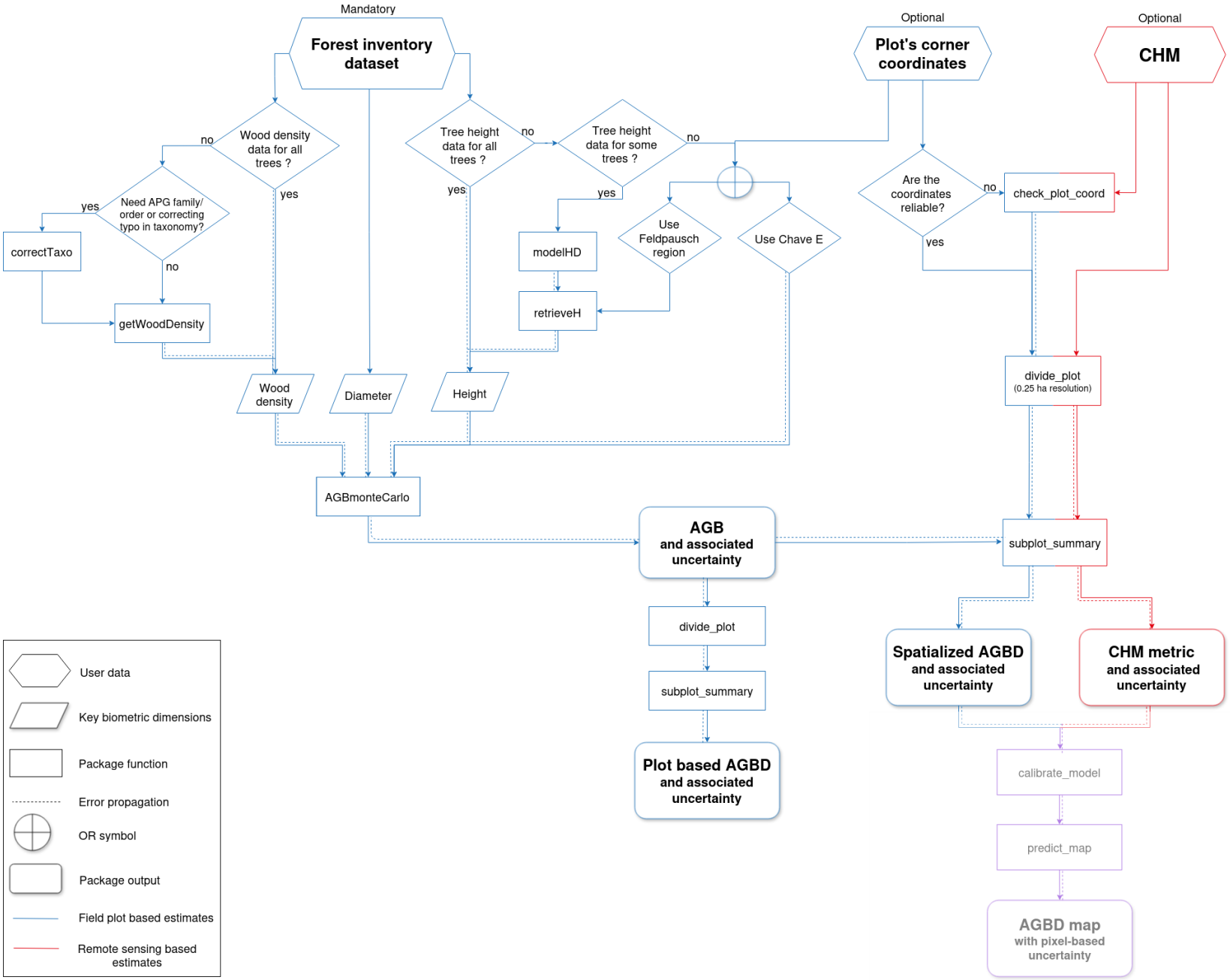


- Tutorial for the entire pipeline divided in three main articles, applied on reduced Nouragues data

Shiny app - BIOMASSapp



- Estimate AGBD per subplot with uncertainties
- Spatialize stand metrics
- No map prediction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BIOMASSapp demonstration

Predict AGBD maps using inventory and LiDAR data

AGBD-LiDAR modelisation: proposed statistical framework

- geostatistical model with SPV-I/C (SPatially Varying Intercept/Coefficients) to integrate spatial correlation:
- $y(s) = (\alpha + \tilde{\alpha}(s)) + (\beta + \tilde{\beta}(s)) \times x(s) + \epsilon(s)$
  with $\tilde{\alpha}(s_1), ..., \tilde{\alpha}(s_n) \sim MVN(0, C_\alpha(s_i, s_j))$
- references

Remote Sensing of Environment 169 (2015) 113–127

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

LiDAR based prediction of forest biomass using hierarchical models with spatially varying coefficients

Chad Babcock [a], Andrew O. Finley [b,*], John B. Bradford [c], Randall Kolka [d], Richard Birdsey [e], Michael G. Ryan [f]

Remote Sensing of Environment 318 (2025) 114557

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

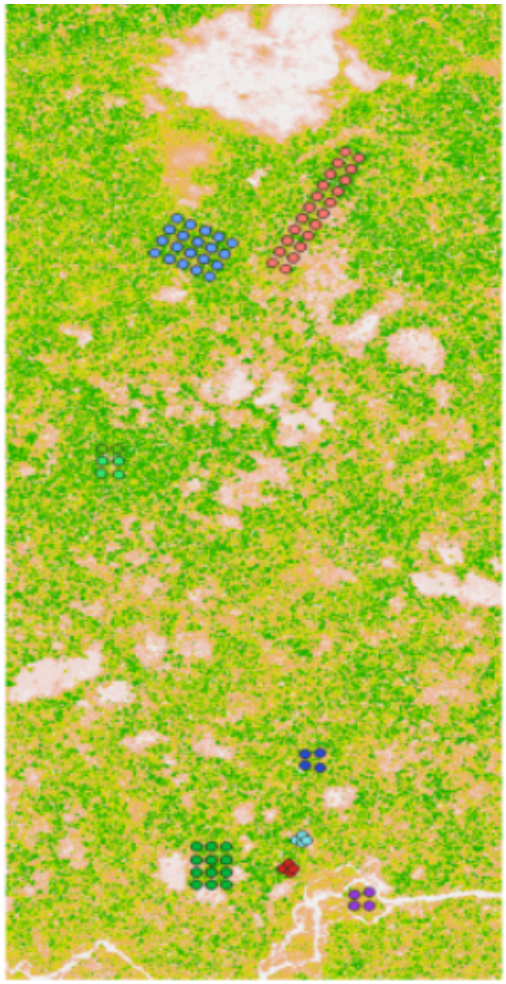
A geostatistical approach to enhancing national forest biomass assessments with Earth Observation to aid climate policy needs

Neha Hunka [a,*], Paul May [b], Chad Babcock [c], José Armando Alanís de la Rosa [d], Maria de los Ángeles Soriano-Luna [d], Rafael Mayorga Saucedo [d], John Armston [a], Maurizio Santoro [e], Daniela Requena Suarez [f], Martin Herold [f], Natalia Málaga [f], Sean P. Healey [g], Robert E. Kennedy [h], Andrew T. Hudak [i], Laura Duncanson [a]

Implementation in BIOMASS

- inference and predictions done using brms R package
- two functions: 1st calibrates the model with user plot and LiDAR data, 2nd predicts AGBD and its associated uncertainties over the full raster footprint provided by the user
- full continuity of the pipeline: the inputs of those functions are the outputs from the previous steps
- propagate all AGBD uncertainties from the previous steps, within Bayesian framework

Nouragues data example



SPV-C model

$log(y(s)) = (\beta + \tilde{\beta}(s)) \times log(x(s)) + \epsilon(s)$

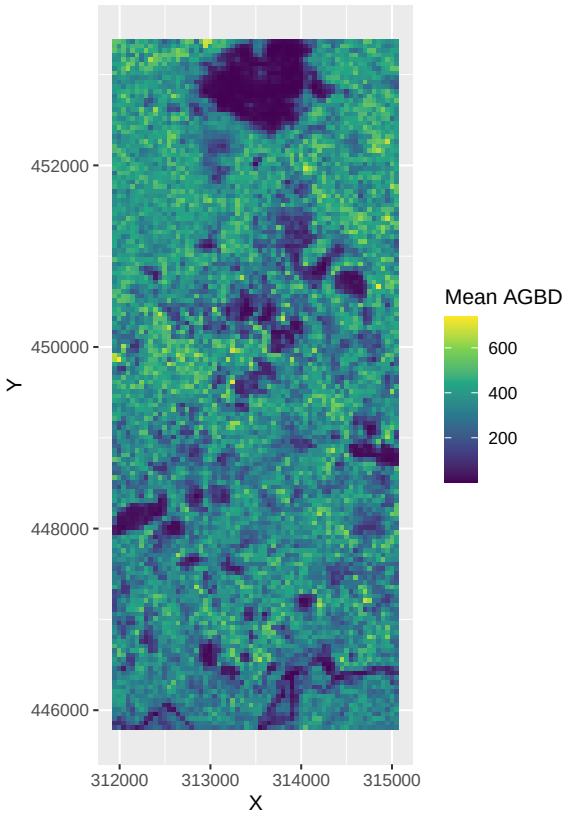
with

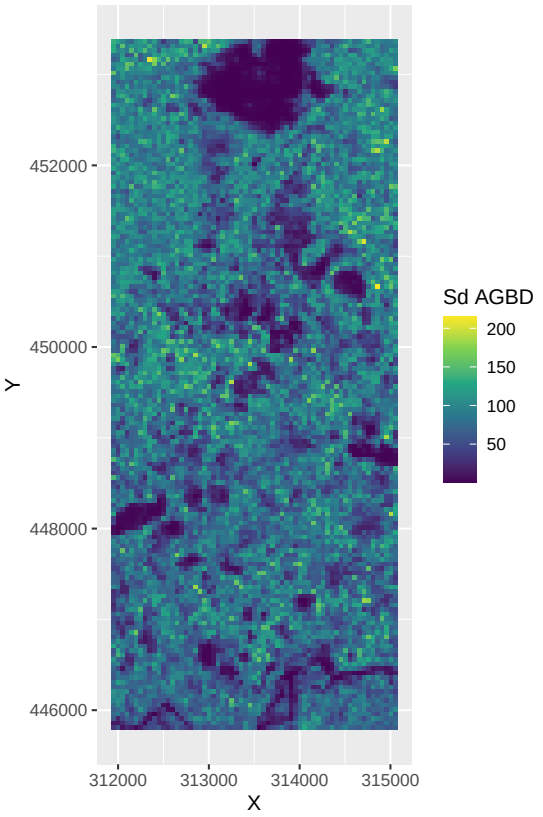
- $y(s)$ AGBD 0.25ha for plot s : ajouter plot de l'agbd carré divisé + incertitudes
- $x(s)$ mean (or median) CHM 0.25ha for plot s : ajouter plot du raster + carré divisé
- $\tilde{\beta}(s_1), ..., \tilde{\beta}(s_n) \sim MVN(0, C_\beta(s_i, s_j))$

Outputs of CHM-AGBD model calibration

ici screen du model_fit + plot ppc

Map predictions





ici nouveau plot avec mean, sd, coeff var (à faire)

Validation

5 What about validation ?

We suggest to validate the model by subsetting your subplot dataset in two: a calibration dataset with about 70% of the subplots, and a validation dataset with the rest. We have 4 plots divided in 16 subplots, so we need to set aside 45 subplots for the calibration step, the rest will be used to validate.

```
# let's select subplots for each step
set.seed(1234)
vector_of_subplots <- unique(dt_inf$subplot_ID)
calibration_subplots <- sample(vector_of_subplots, 45)
validation_subplots <- vector_of_subplots[!(vector_of_subplots %in% calibration_subplots)]

# then divide our complete dataset
dt_calibration <- dt_inf %>%
  filter(subplot_ID %in% calibration_subplots)
dt_validation <- dt_inf %>%
  filter(subplot_ID %in% validation_subplots) %>%
  # change in column names to make predict function work
  mutate(x = x_center) %>%
  mutate(y = y_center) %>%
  mutate(log_CHM = log(raster_metric))
```

Now we can calibrate the model (ie run the inference) with the calibration dataset:

```
cal_step1 <- calibrate_model(long_AGB_simu = dt_calibration, nb_rep = 50, useCache = T,
                             plot_model = TRUE, chains = 4, thin = 20, iter = 2500,
                             warmup = 500, cores = 4)
```

Now we are going to predict the model outputs (namely AGBD) at observed data points using the parameters that we inferred at step 1 when calibrating. For that we use the predict function of brms (link to help):

```
val_step2 <- predict(cal_step1, newdata = dt_validation, ndraws = 1)
```

Now we may compare those values predicted by the model, which was calibrated with the calibration dataset, with observed values from the validation dataset:

```
dt_validation <- cbind(dt_validation, logAGBD_pred = val_step2[,1])

ggplot(dt_validation, aes( x = log(AGBD), y = logAGBD_pred, colour = subplot_ID))+
  geom_point(shape = 1)+
  geom_abline(colour = "red", linetype = 2, linewidth = 1.2)+
  xlab("Observed log AGBD")+
  ylab("Predicted log AGBD")
```

In the vignette we propose a example of validation method.

Future work, this week and later on

- More robust quality check with single or multiple dates
- New generic global allometry for AGB
- Implementation of one or more new approaches for estimating height in the absence of field data
- Taking into account positioning uncertainty on individual trees
- Management of circular plots in the spatialisation pipeline for estimates
- Flexibility of the AGBD-LiDAR model (implemented in v3.0)
- Temporal BIOMASS

Authors and funders

Maxime Réjou-Méchain, Arthur Bailly, Guillaume Cornu, John Godlee, Fabian Fischer, Jérome Chave, Arthur Pere, Ariane Tanguy, Camille Piponiot, Bruno Hérault, Philippe Verley, Ted Feldpausch

European Spatial Agency

Research Institute for Development

the French agricultural research and cooperation organization working for the sustainable development of tropical and Mediterranean regions